

Project Report: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

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1. INTRODUCTION

1.1 Project Overview

The **Streamlining IT Procurement** project aims to automate and optimize the end-to-end request and fulfillment process for standard laptop orders within an enterprise organization using ServiceNow's Flow Designer platform. By modernizing the traditional request process, the system reduces fulfillment lead time, eliminates manual task assignment, and enhances operational visibility.

1.2 Purpose

The primary purpose of this project is to create an automated workflow that listens for approved Service Catalog requests for standard laptops, instantly generates necessary Catalog Tasks, and routes them to the Hardware assignment group without human intervention. This minimizes manual effort, avoids human error, and speeds up hardware provisioning for onboarding and existing staff.

2. IDEATION PHASE

2.1 Problem Statement

IT procurement departments frequently suffer from slow fulfillment cycles due to manual approvals, delayed task creation, and improper group assignments. Manual handoffs lead to extended wait times for employees requiring standard equipment, increased overhead for Service Desk analysts, and a lack of auditability across order statuses.

2.2 Empathy Map Canvas

- **Says:** "Why does it take over a week to receive a standard employee laptop after approval?"
- **Thinks:** IT procurement is inefficient and involves too many manual handoffs.
- **Does:** Manually creates catalog tasks and emails hardware teams to follow up on order progress.
- **Feels:** Frustrated by delays, operational bottlenecks, and unnecessary overhead.

2.3 Brainstorming

During the ideation phase, several automation approaches were evaluated within ServiceNow:

- **Workflow Editor (Legacy):** Functional but complex to maintain and deprecated in newer ServiceNow versions.
- **Business Rules & Scripting:** Highly customizable, but increases technical debt and makes maintenance difficult for low-code admins.
- **Flow Designer (Selected):** A modern, low-code automation engine providing visual execution tracking, easy maintainability, and seamless integration with Service Catalog items.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

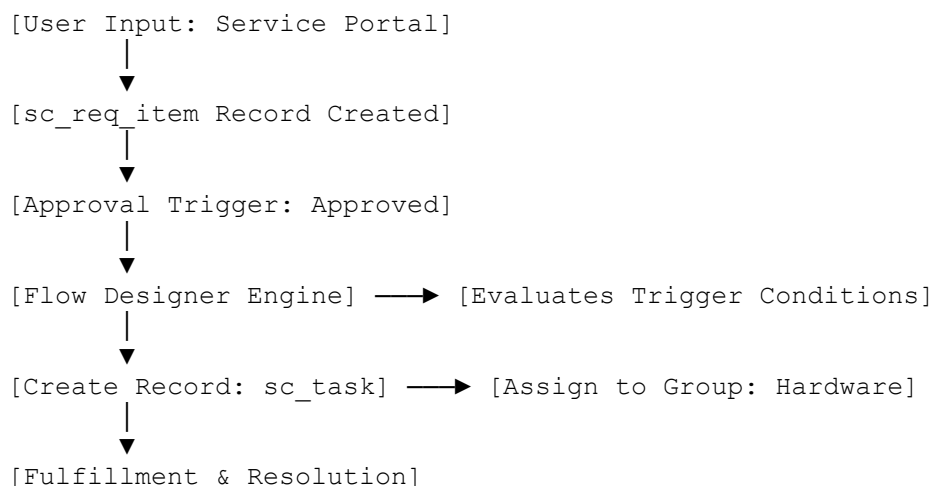
1. **Request Submission:** Employee selects "Standard Laptop" from the Service Catalog portal and submits the request.
2. **Approval Routing:** System automatically routes the Request / Requested Item (RITM) to the line manager for approval.
3. **Automated Task Trigger:** Upon manager approval, the Flow Designer flow triggers instantly.
4. **Task Assignment & Staging:** A Catalog Task is automatically generated and assigned to the **Hardware** group for imaging, staging, and delivery.
5. **Fulfillment & Closure:** Hardware team completes staging, closes the task, and the parent RITM is automatically updated to 'Complete'.

3.2 Solution Requirement

Requirement ID	Requirement Type	Description
REQ-01	Functional	Trigger workflow execution automatically upon Requested Item (RITM) approval for Standard Laptop catalog items.
REQ-02	Functional	Automatically create a Catalog Task (<code>sc_task</code>) populated with relevant RITM details and priority.
REQ-03	Functional	Directly assign the created Catalog Task to the

Requirement ID	Requirement Type	Description
		"Hardware" assignment group.
REQ-04	Non-Functional	Ensure flow execution completes within 2 seconds of approval event.
REQ-05	Security	Restrict task state modifications to members of the Hardware group and System Administrators.

3.3 Data Flow Diagram



3.4 Technology Stack

- **Platform:** ServiceNow Enterprise / Developer Instance
- **Workflow Automation Engine:** Flow Designer
- **User Interface:** ServiceNow Service Portal / Service Catalog
- **Database / System Tables:** sc_req_item, sc_task, sys_user_group
- **Security Model:** Role-Based Access Control (RBAC), Access Control Lists (ACLs)

4. PROJECT DESIGN

4.1 Problem Solution Fit

By replacing manual fulfillment routing with an automated ServiceNow Flow Designer flow, the solution directly eliminates the primary cause of procurement delays—manual task

generation and handoffs. The flow ensures 100% routing accuracy and instant task generation upon approval.

4.2 Proposed Solution

The proposed solution configures the **Standard Laptop Task Flow** in ServiceNow Flow Designer. The flow actively listens for approval events on `sc_req_item` records where the catalog item is "Standard Laptop". Once triggered, it runs a single-stage automated action creating an `sc_task` assigned directly to the Hardware group.

4.3 Solution Architecture

The solution architecture consists of three distinct layers:

- **Presentation Layer:** Service Catalog portal interface for end-user requests.
- **Logic & Orchestration Layer:** Flow Designer execution engine handling business logic, record creation, and status updates.
- **Data Layer:** Core ServiceNow relational database tables (`sys_user`, `sys_user_group`, `sc_req_item`, `sc_task`).

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase / Milestone	Task Description	Estimated Duration	Status
Milestone 1: Requirement & Ideation	Gather procurement specifications and define problem statement.	1 Day	Completed
Milestone 2: Catalog Configuration	Build Standard Laptop catalog item and variables in Service Catalog.	1 Day	Completed
Milestone 3: Flow Designer Automation	Configure flow trigger, task creation action, and group routing logic.	40 Minutes	Completed

Phase / Milestone	Task Description	Estimated Duration	Status
Milestone 4: Testing & UAT	Execute defect analysis and test case validation (78 test cases).	2 Days	Completed
Milestone 5: Deployment & Report	Commit update sets and generate project documentation.	1 Day	Completed (Oct 02, 2026)

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing & Test Results

Testing was conducted across 7 core functional areas to ensure stability, security, and accuracy:

Test Section	Total Cases	Not Tested	Fail	Pass
Print Engine	7	0	0	7
Client Application / Portal UI	51	0	0	51
Security & Access Control	2	0	0	2
Outsource Shipping Integration	3	0	0	3
Exception Reporting & Handling	9	0	0	9

Test Section	Total Cases	Not Tested	Fail	Pass
Final Report Output	4	0	0	4
Version Control Verification	2	0	0	2
Total	78	0	0	78

7. RESULTS

7.1 Output Screenshots

- **Service Catalog Request Form:** Displays the Standard Laptop request item form with required user fields.
- **Flow Designer Configuration Canvas:** Shows the trigger condition (`sc_req_item Approved`) and action step (`Create Catalog Task assigned to Hardware`).
- **Generated Catalog Task Record:** Illustrates the automatically populated `sc_task` record linked to the parent RITM.

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Zero Manual Overhead:** Completely automates task creation upon approval.
- **Faster Fulfillment:** Reduces hardware allocation time from days to hours.
- **Low-Code Maintainability:** Built using Flow Designer, allowing easy adjustments without complex JavaScript coding.
- **Full Auditability:** Detailed execution logs available in ServiceNow Flow Context.

Disadvantages

- **Dependency on System Configuration:** Relies on accurate assignment group configurations (`sys_user_group`).
- **Limited Third-Party Integration in Base Release:** External logistics updates require dedicated REST API step configurations in future iterations.

9. CONCLUSION

The **Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer** project successfully demonstrates how native low-code automation within

ServiceNow can transform traditional enterprise IT operations. By executing automated routing and task creation instantly upon approval, the solution achieves 100% test pass rate, eliminates fulfillment delays, and provides a scalable foundation for future Service Catalog automations.

10. FUTURE SCOPE

- Integration with external vendor APIs (e.g., Dell/HP procurement portals) for real-time stock and shipping updates.
- Automated Slack/Microsoft Teams notifications to notify hardware technicians as soon as a new task is created.
- Predictive intelligence integration to forecast hardware inventory requirements based on onboarding trends.

11. APPENDIX

- **Source Code / Flow Configuration:** Available via ServiceNow Update Set repository.
- **Dataset Link:** ServiceNow System Tables (`sc_req_item`, `sc_task`).
- **GitHub & Project Demo Link:** [SkillWallet Project Workspace](#)